LECTURE 16 · DISCRETE MATHEMATICS

Order without a single ranking

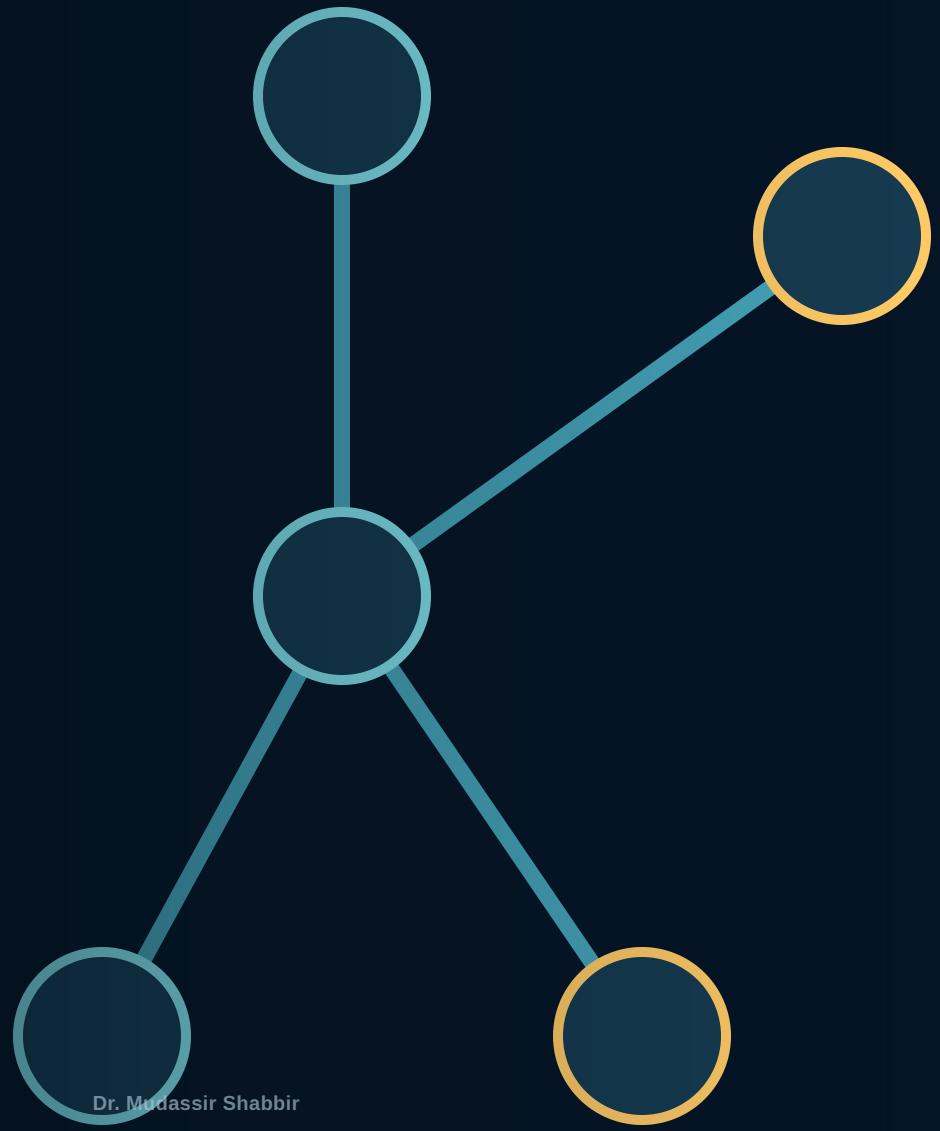
Can 8 and 12 both sit at the top?



01 · PARTIAL ORDERS

Which relations create order?

Keep reflexivity and transitivity. Require antisymmetry.



Three properties define a partial order

A poset is a set together with a partial order: $(S, \preceq)$.

Reflexive

$a \preceq a$ **for every** $a \in S$**.**

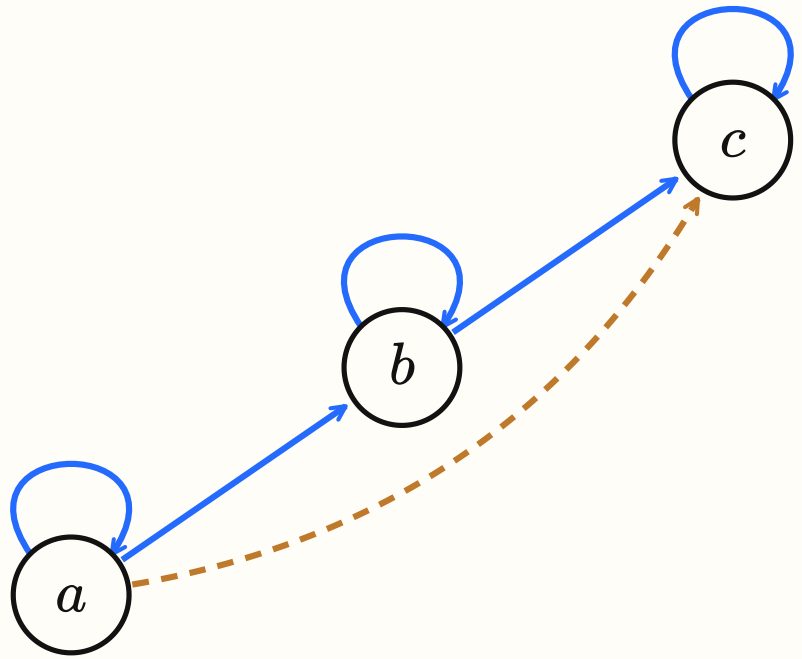
Antisymmetric

$a \preceq b \land b \preceq a \implies a = b$**.**

Transitive

$a \preceq b \land b \preceq c \implies a \preceq c$**.**

A three-element partial order



Loops: reflexive. Dashed arrow: transitive. No opposite arrows between distinct vertices.

$\preceq$ names the relation; it need not mean numerical $\leq$.

EXERCISE 1

Does “greater than or equal” give a poset?

Test $(\mathbb{Z}, \geq)$ against all three properties.

01 **Is it reflexive?**

Yes: $a \geq a$.

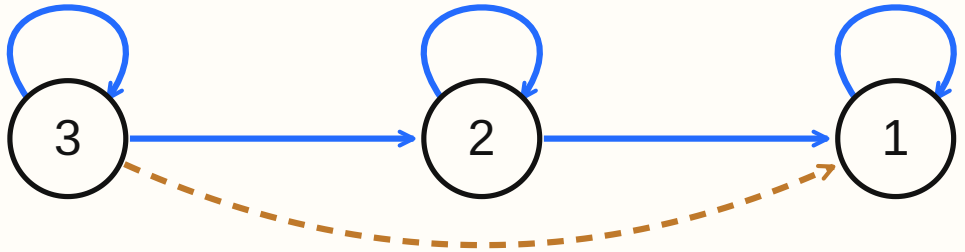
02 **Is it antisymmetric?**

Yes: $a \geq b$ **and** $b \geq a$ **force** $a = b$.

03 **Is it transitive?**

Yes: $a \geq b$ **and** $b \geq c$ **give** $a \geq c$.

A finite sample of $(\mathbb{Z}, \geq)$



An arrow $a \to b$ means $a \geq b$.

The direction of an order is part of its definition.

Divisibility orders the positive integers

Write $a \preceq b$ when $a \mid b$.

Reflexive

$a = a \cdot 1$, **so** $a \mid a$**.**

Antisymmetric

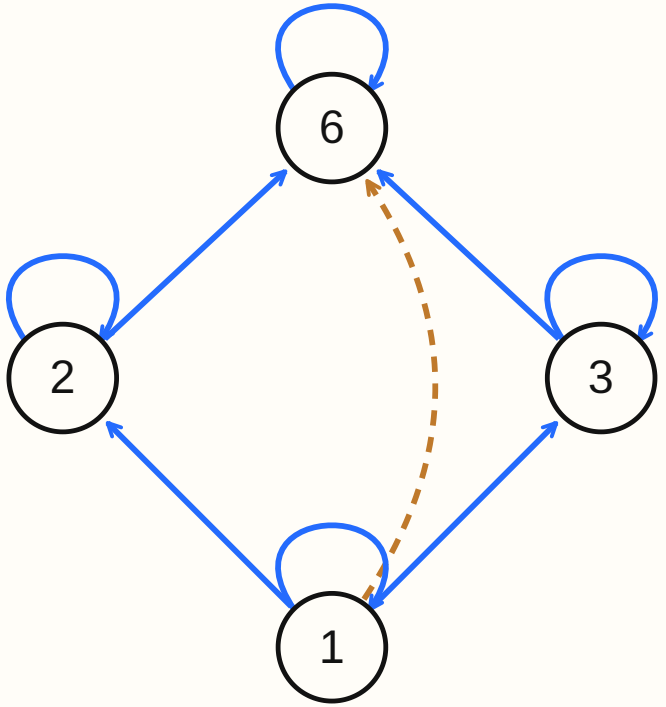
$b = ak,\ a = b\ell \implies k\ell = 1$

For positive integers, $k = \ell = 1$; hence $a = b$.

Transitive

$b = ak,\ c = b\ell \implies c = a(k\ell)$**.**

A finite sample: $\{1, 2, 3, 6\}$



Every arrow means “divides”; all four loops are included.

Containment orders subsets

On $\mathcal{P}(U)$, use $A \preceq B$ when $A \subseteq B$.

Reflexive

$$A \subseteq A.$$

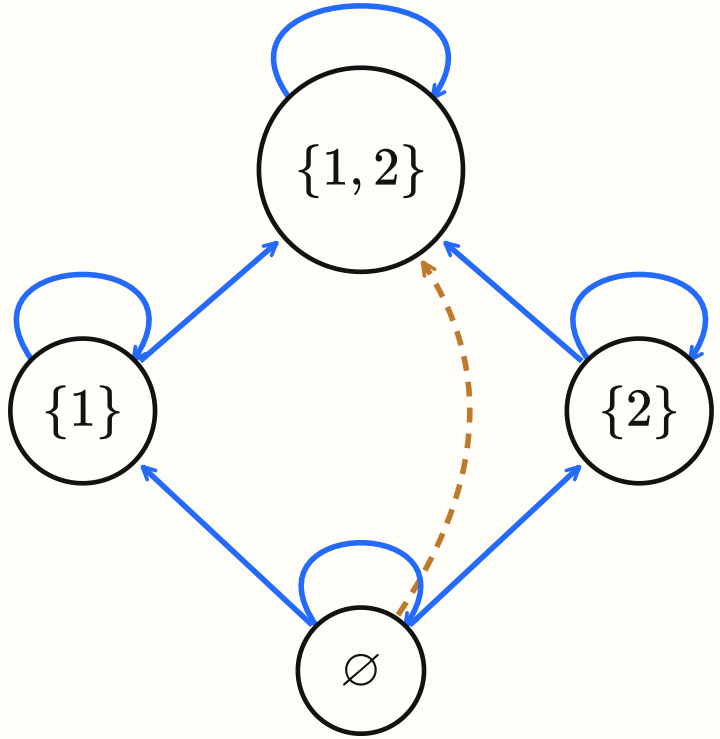
Antisymmetric

$$A \subseteq B \ \wedge \ B \subseteq A \implies A = B.$$

Transitive

$$A \subseteq B \ \wedge \ B \subseteq C \implies A \subseteq C.$$

$\mathcal{P}(\{1,2\})$, **ordered by** $\subseteq$



The singletons are incomparable. The dashed arrow follows by transitivity.

Every example uses the same three checks.

EXERCISE 2

One failed property is enough

Which relations are partial orders on the stated sets?

01 $<$ **on** $\mathbb{Z}$

No: $a < a$ is false, so reflexivity fails.

02 **Divisibility on all of** $\mathbb{Z}$

No: $2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$.

03 **Congruence modulo 3 on** $\mathbb{Z}$

No: $1 \equiv 4$ and $4 \equiv 1 \pmod 3$, but $1 \neq 4$.

$<$ on $\mathbb{Z}$



Missing loops violate reflexivity.

Divisibility on $\mathbb{Z}$



Two distinct elements divide each other.

Congruence modulo 3



Distinct elements are related both ways.

EXERCISE 3

Comparable means the relation holds one way

a, b are comparable when $a \preceq b$ or $b \preceq a$. Otherwise they are incomparable.

01 **In $(\mathbb{Z}^+, |)$: compare 3 and 9.**

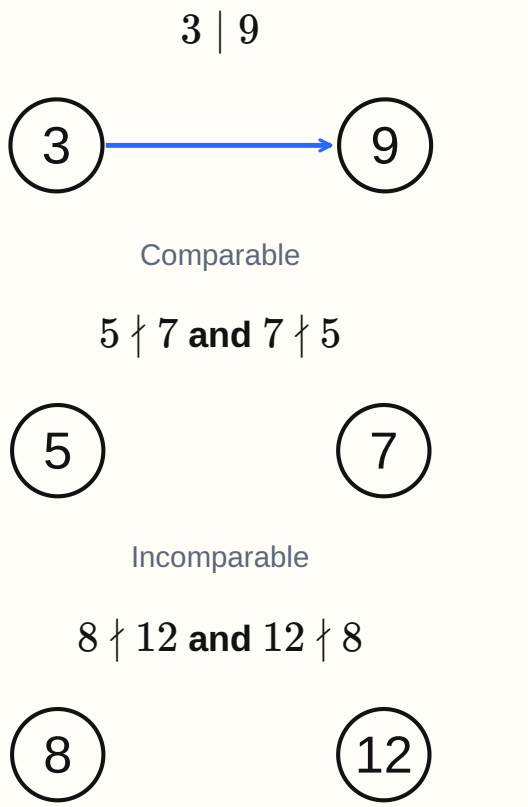
Comparable: $3 \mid 9$.

02 **Compare 5 and 7.**

Incomparable: $5 \nmid 7$ and $7 \nmid 5$.

03 **Compare 8 and 12.**

Incomparable, although $8 < 12$ numerically.



A total order compares every pair

A total (linear) order is a partial order with no incomparable pair. A totally ordered set is also called a chain.

Total order

$(\mathbb{Z}, \leq)$

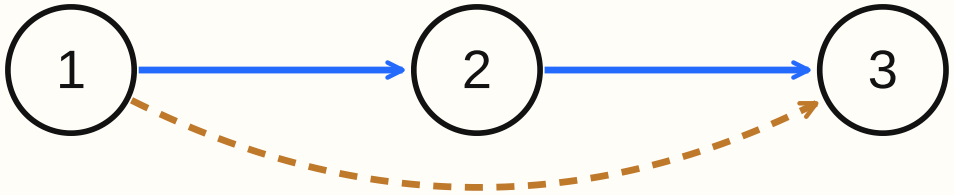
For every a, b, either $a \leq b$ or $b \leq a$.

Partial, but not total

$(\mathbb{Z}^+, |)$

5 and 7 provide a counterexample to totality.

Total: every pair is comparable



A finite sample of $(\mathbb{Z}, \leq)$; loops omitted.

Partial: a pair can be incomparable



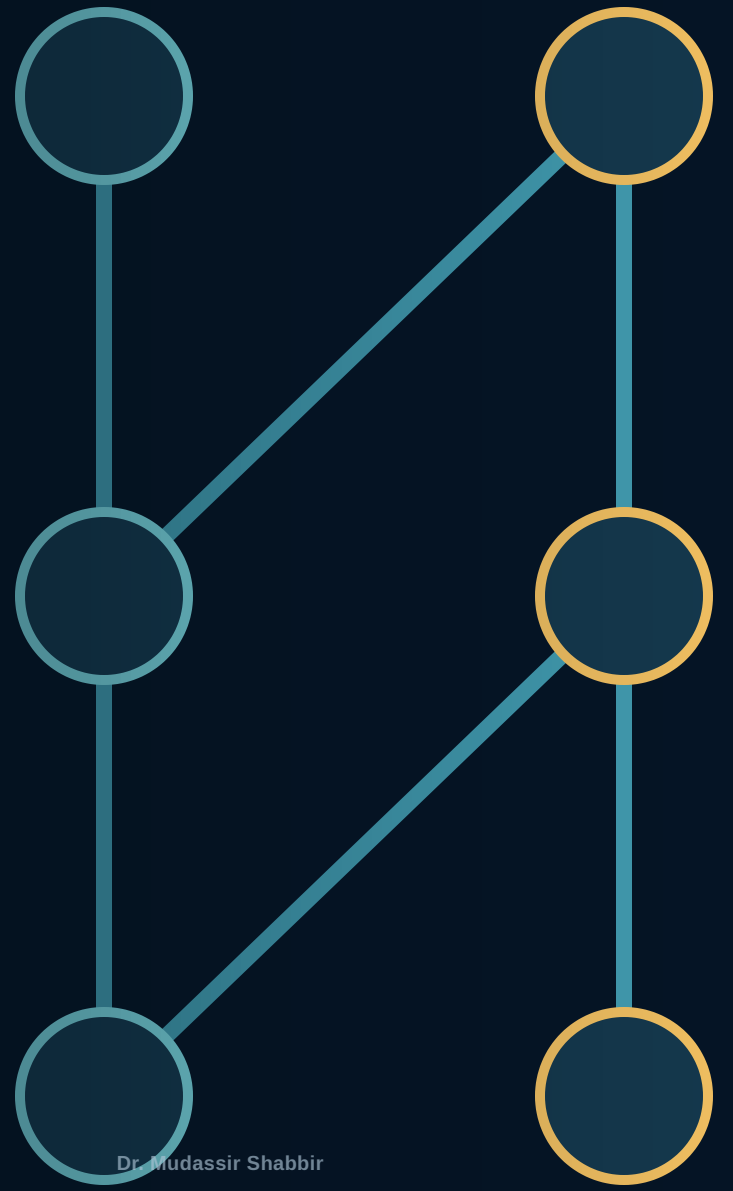
A finite sample of $(\mathbb{Z}^+, |)$; loops omitted.

“Partial” allows incomparability; it does not require it.

02 · HASSE DIAGRAMS

Draw only the essential edges

A Hasse diagram stores the comparisons that force all the others.



A cover has no intermediate element

Write $x \prec y$ for $x \preceq y$ and $x \neq y$.

Keep the cover

y covers x if $x \prec y$ and no z satisfies $x \prec z \prec y$.

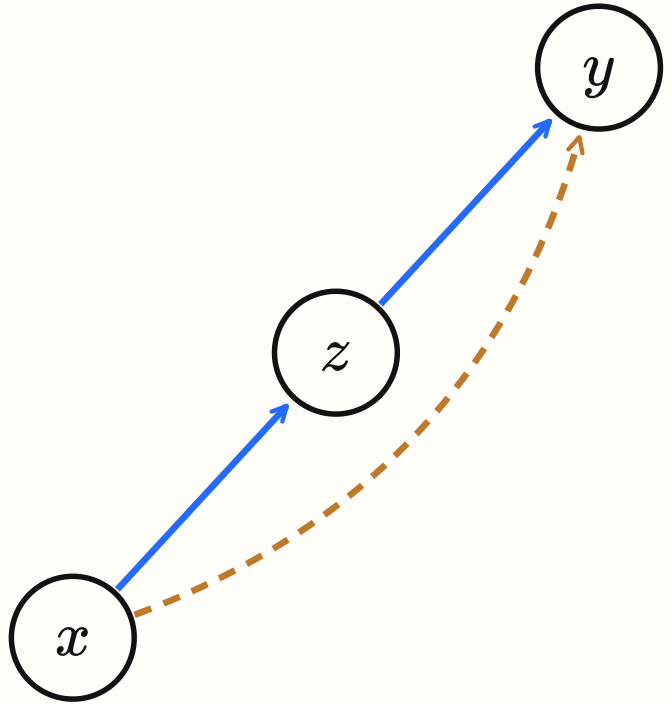
Remove only what is implied

Omit loops and transitive shortcuts; put the greater endpoint above.

Read upward

$x \preceq y$ means $x = y$ or an upward path leads from x to y.

An intermediate element rules out a cover



$x \prec z \prec y$: y does not cover x. The solid edges are covers.

For finite posets, covers determine the entire relation.

Constructing a Hasse diagram

$S = \{1, 2, 3, 4, 6, 8, 12\}$, ordered by divisibility. Begin with $a \to b$ whenever $a \mid b$.

00 Arrange vertices by level

Place each greater endpoint above its smaller endpoint.

01 Remove the loops

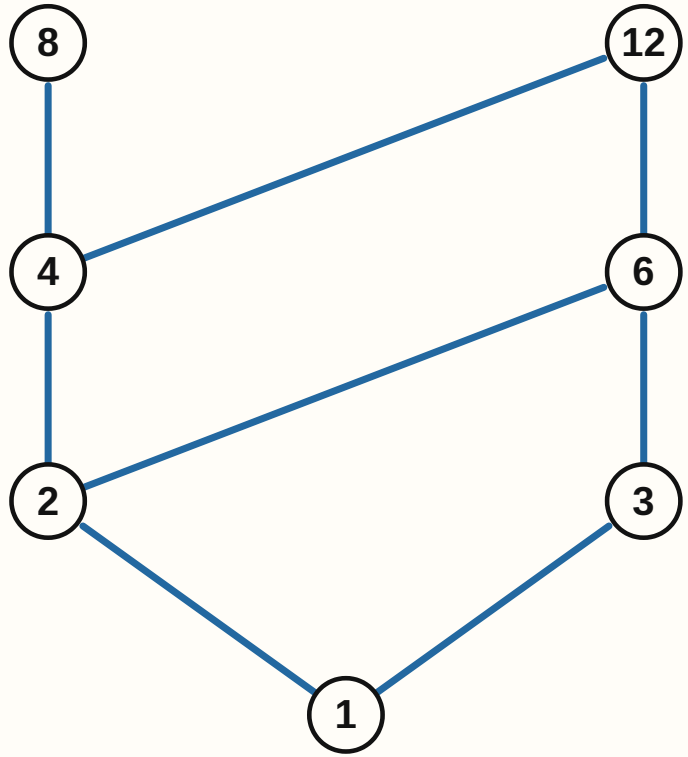
Every element still divides itself.

02 Remove transitive edges

An upward path already implies the comparison.

03 Keep covers; omit arrowheads

Keep eight covers; read every edge upward.



Hasse diagram · 8 cover edges

The relation stays the same: $a \mid b$ means $a = b$ or an upward path from a to b.

EXERCISE 4

An upward path is a comparison

Use the Hasse diagram of $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.

01 **Is $2 \preceq 12$ even without an edge?**

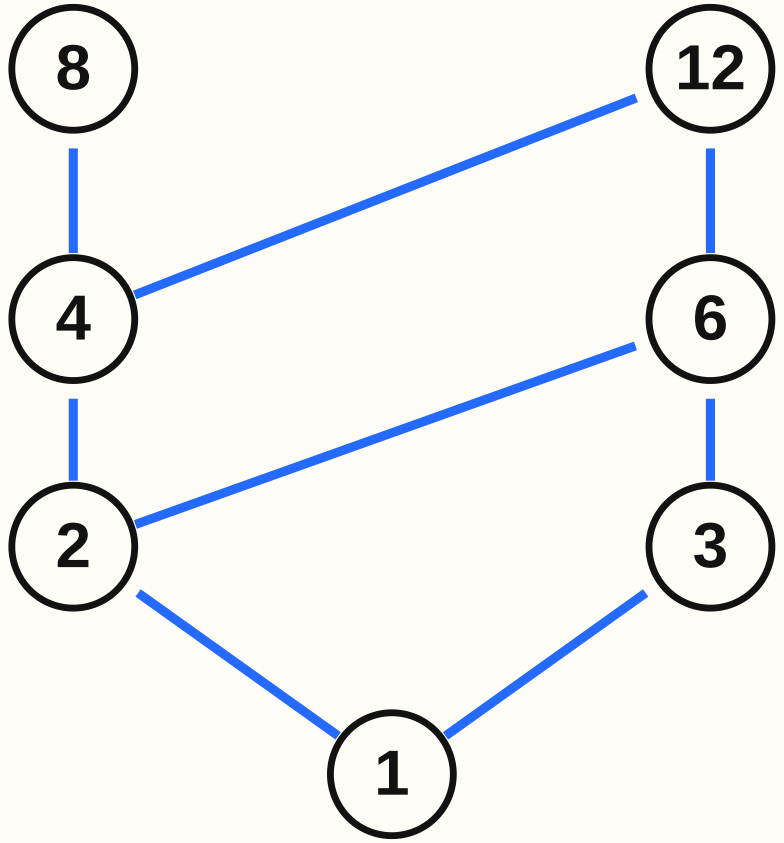
Yes: $2 \to 4 \to 12$ (also $2 \to 6 \to 12$).

02 **Are 3 and 4 comparable?**

No upward path connects them in either direction.

03 **Does being drawn higher make 8 greater than 6?**

No. Height alone is insufficient: $6 \nmid 8$ and $8 \nmid 6$.



EXERCISE 5

Build a Hasse diagram from scratch

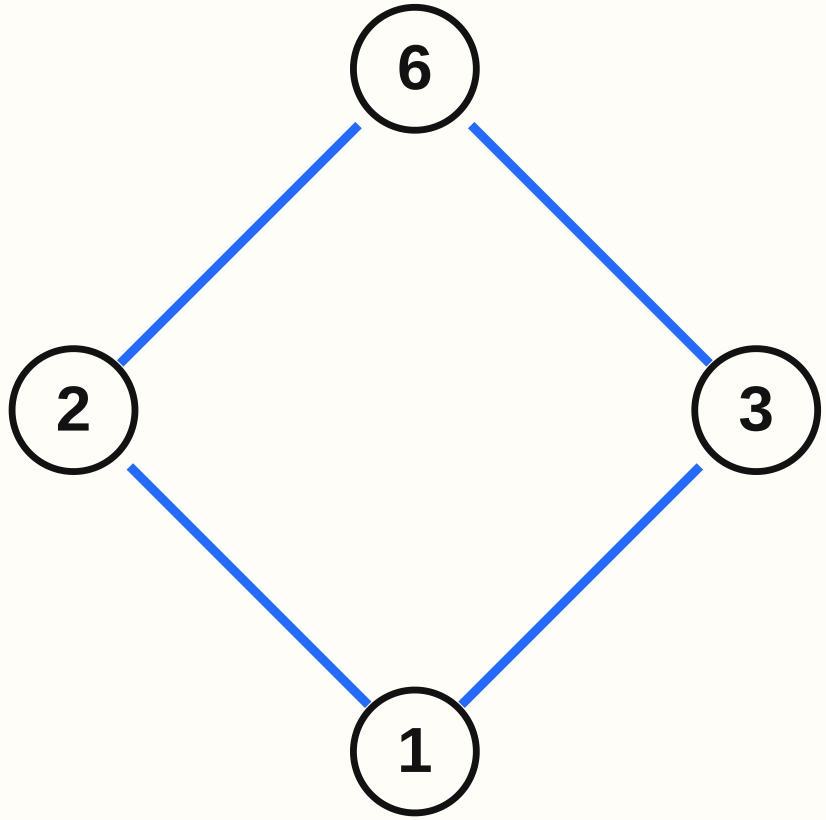
For $S = \{1, 2, 3, 6\}$ ordered by divisibility:

01 **Which pairs are covers?**

$(1, 2), (1, 3), (2, 6), (3, 6)$.

02 **Should you draw an edge from 1 to 6?**

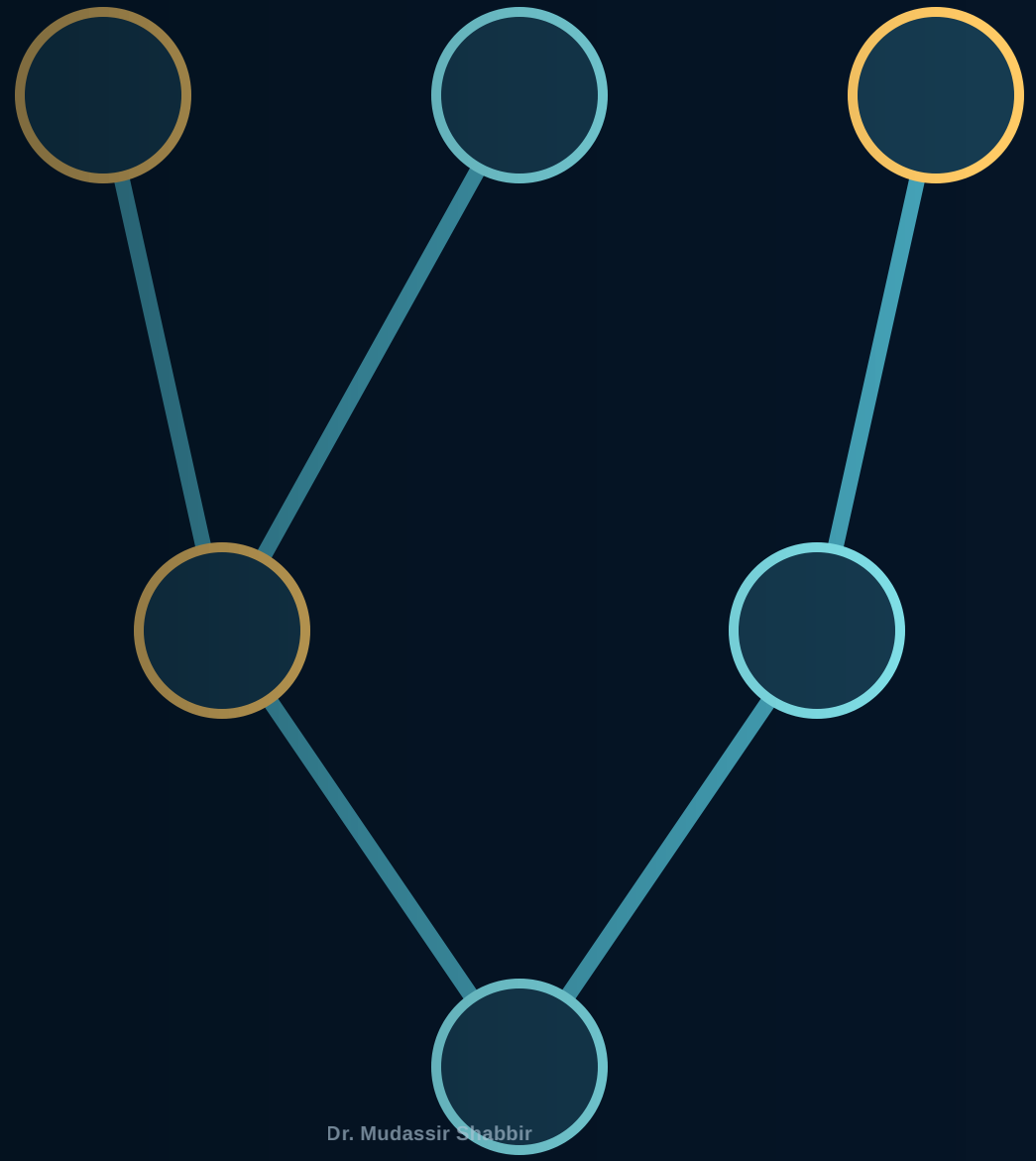
No: there is an intermediate element, 2 (or 3).



03 · EXTREMA

At the top of which order?

An element can be maximal without being greatest.



Minimal and maximal are local conditions

Minimal

No $y \neq x$ satisfies $y \preceq x$.

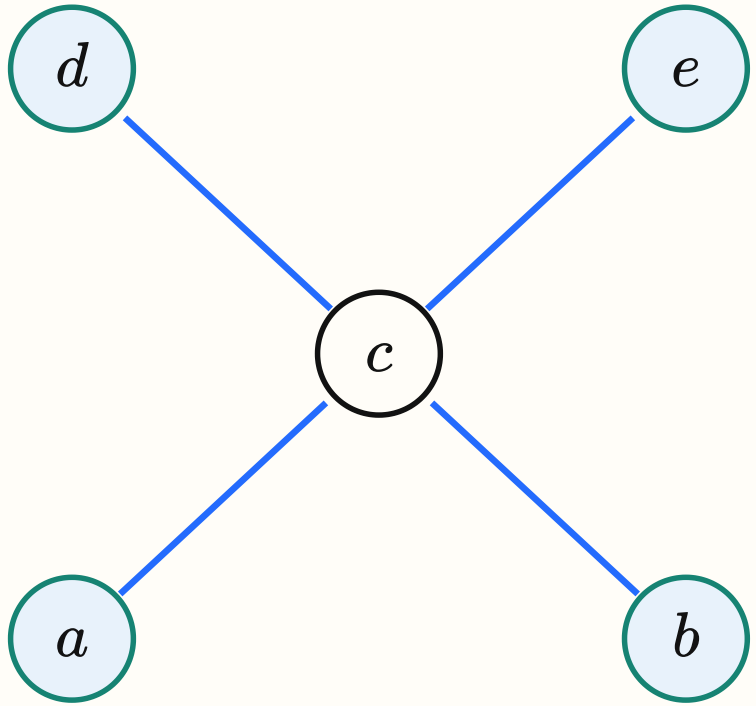
No element sits strictly below x in the order.

Maximal

No $y \neq x$ satisfies $x \preceq y$.

No element sits strictly above x in the order.

Several extrema can coexist



Minimal: a, b. Maximal: d, e. Neither a least nor a greatest element.

A finite diagram: minimal vertices have no edge down; maximal vertices have no edge up.

EXERCISE 6

Find every minimal and maximal element

Work in $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.

01 **Which elements are minimal?**

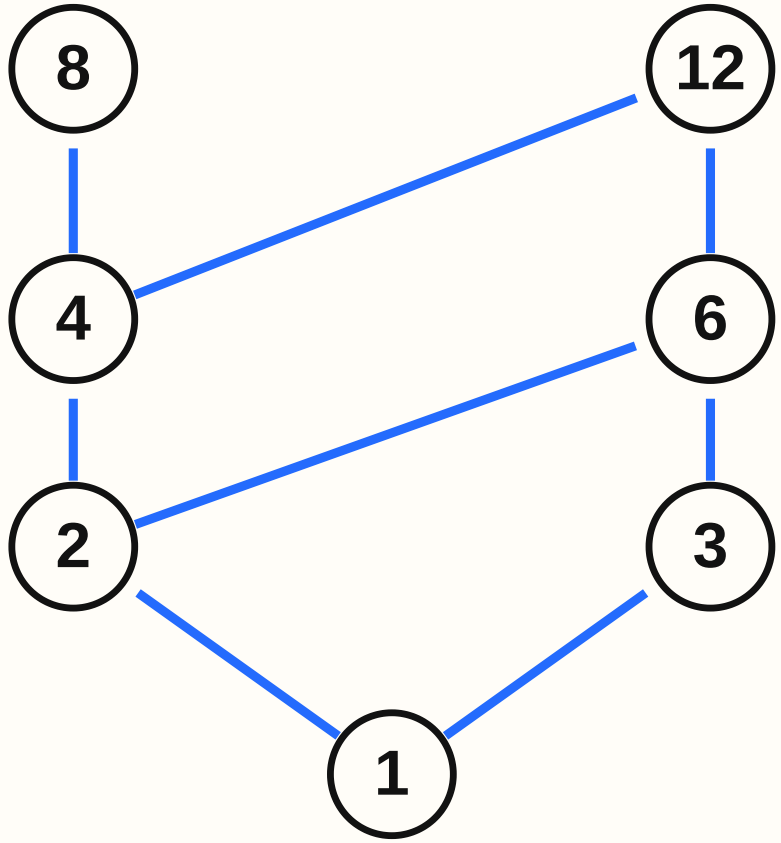
Only 1: every other vertex has something below it.

02 **Which elements are maximal?**

8 and 12: neither has an upward edge.

03 **Why is 8 still maximal when $8 < 12$?**

Maximality uses divisibility, and $8 \nmid 12$.



Least and greatest compare with everyone

Least element

$\ell \preceq y$ **for every** $y \in S$**.**

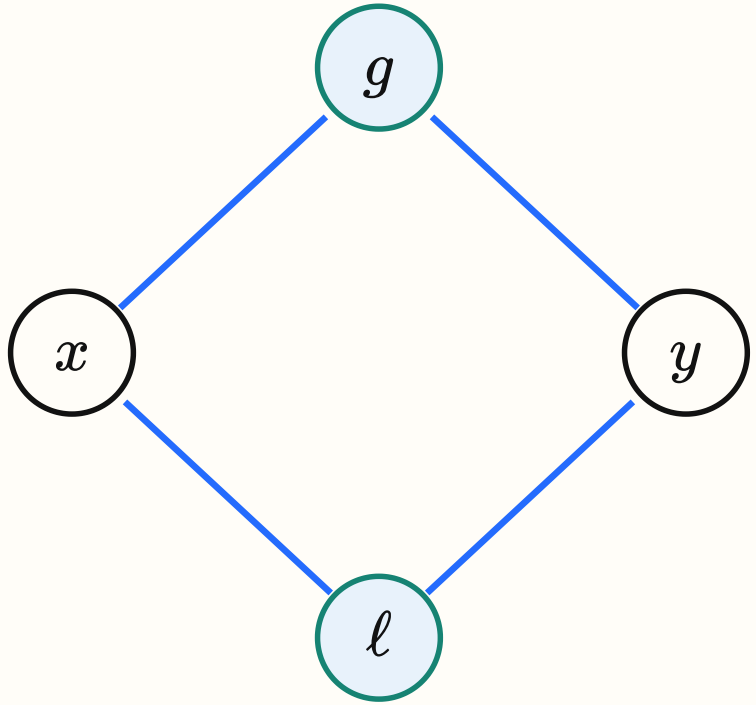
A least element lies below every element.

Greatest element

$y \preceq g$ **for every** $y \in S$**.**

A greatest element lies above every element.

Reach every vertex by an upward path



ℓ is below everyone; g is above everyone.

Least implies minimal. Greatest implies maximal. The converses fail in general.

EXERCISE 7

Two maximal elements; no greatest element

Use the same divisibility poset.

01 **Does a least element exist?**

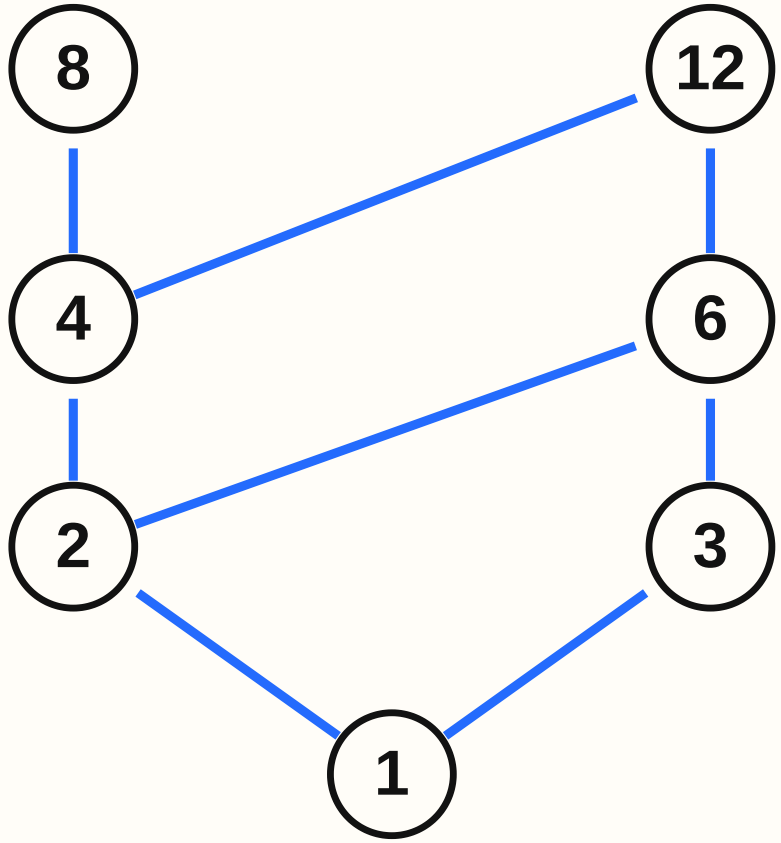
Yes: 1 divides every element of S.

02 **Does a greatest element exist?**

No: neither 8 nor 12 is divisible by the other.

03 **Could 24 be the greatest element here?**

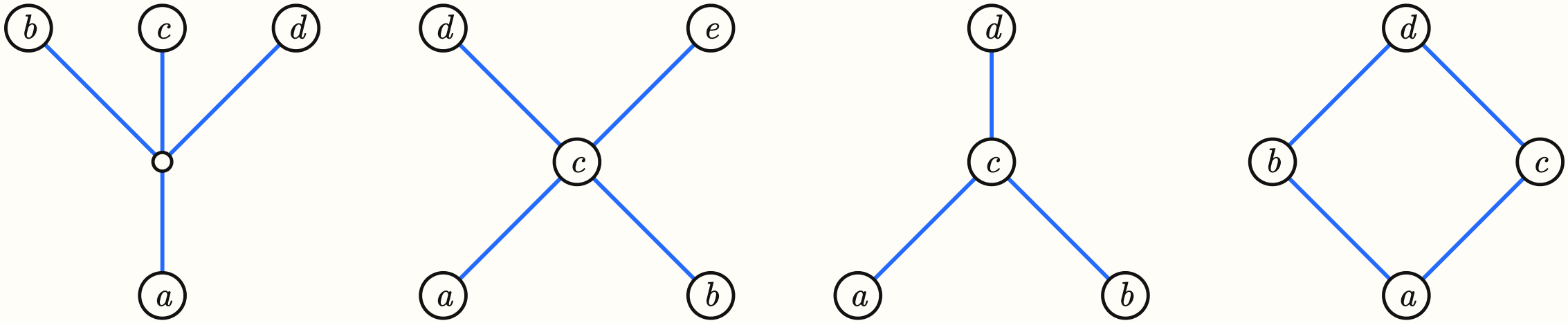
No: $24 \notin S$. Extrema must belong to the poset.



The numerically largest member need not be greatest in this order.

EXERCISE 8

Read four different outcomes



I II III IV

01 **Which diagrams have a greatest element?**

None; none; d; d.

02 **Which diagrams have a least element?**

a; none; none; a.

A greatest element cannot have a rival

Suppose g and h are both greatest in the same poset.

Use the definition twice

$g \preceq h$ because h is greatest; $h \preceq g$ because g is greatest.

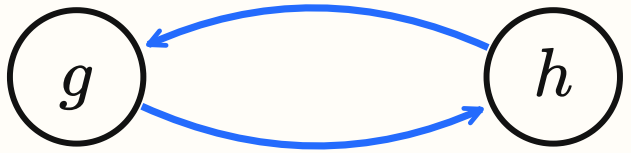
Use antisymmetry

$g = h$.

Reverse the argument

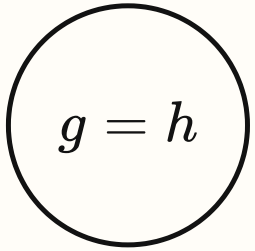
A least element is also unique, when it exists.

Suppose both are greatest



$g \preceq h$ and $h \preceq g$.

Antisymmetry identifies them



They are the same element, not two rivals.

Several maximal elements are possible; several greatest elements are not.

The underlying set changes the answer

All subsets, ordered by containment

$(\mathcal{P}(U), \subseteq)$

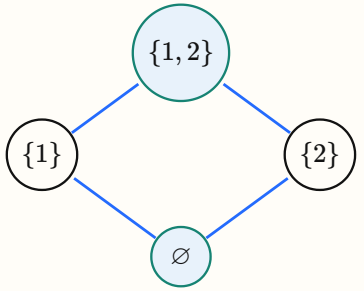
Least: $\varnothing$. Greatest: U.

Positive integers, ordered by divisibility

$(\mathbb{Z}^+, |)$

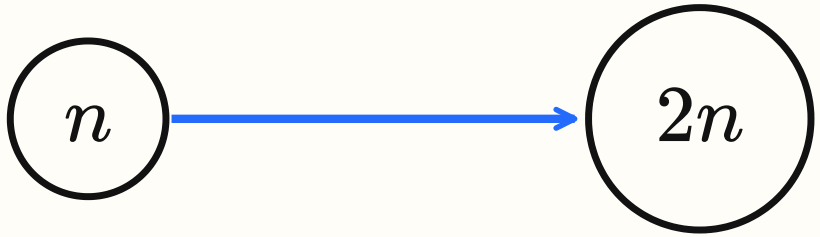
Least: 1. No greatest: $n \mid 2n$ and $n \neq 2n$.

Example: $U = \{1, 2\}$



$\varnothing$ is least; U is greatest.

Every positive integer has a successor



$n \mid 2n$ and $n \neq 2n$; no greatest element.

Always specify both the set and the relation.

EXERCISE 9

Infinite orders may have no extrema

Compare these familiar posets.

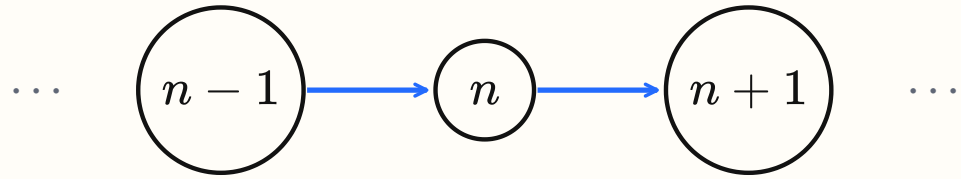
01 **In $(\mathbb{Z}, \leq)$, find all minimal and maximal elements.**

None: for each n, $n-1 < n < n+1$.

02 **In $(\mathbb{Z}^+, |)$, find all maximal elements.**

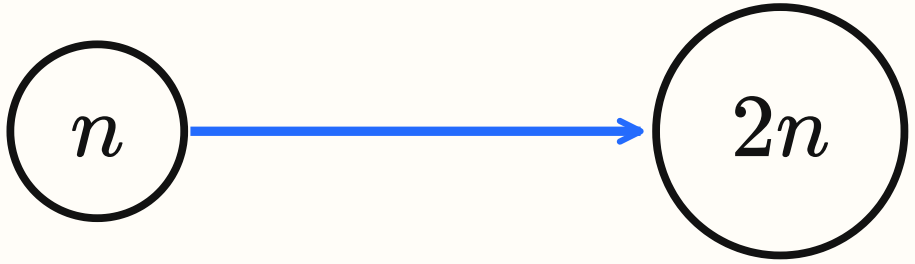
None: every n has the strictly greater element $2n$.

$(\mathbb{Z}, \leq)$ continues both ways



For every n, there is a smaller and a larger integer.

$(\mathbb{Z}^+, |)$ always extends



Every n has a strictly greater element $2n$.

EXERCISE 10 · OPTIONAL CHALLENGE

How large can an intersecting family be?

Let $n \geq 1$. Distinct subsets $A_1, \ldots, A_k \subseteq U = \{1, \ldots, n\}$ satisfy $A_i \cap A_j \neq \varnothing$ for every i, j.

01 **Show that $k \leq \ldots$**

Pair each subset with its complement. At most one member of each pair can be selected.

02 **Give a family attaining equality.**

All subsets containing 1: there are 2^{n-1} of them.

OPTIONAL CHALLENGE · SOLUTION

Complement pairs force the bound

For $n \geq 1$, a subset never equals its own complement.

Count the pairs

2^n subsets form 2^{n-1} pairs $\{A, U \setminus A\}$.

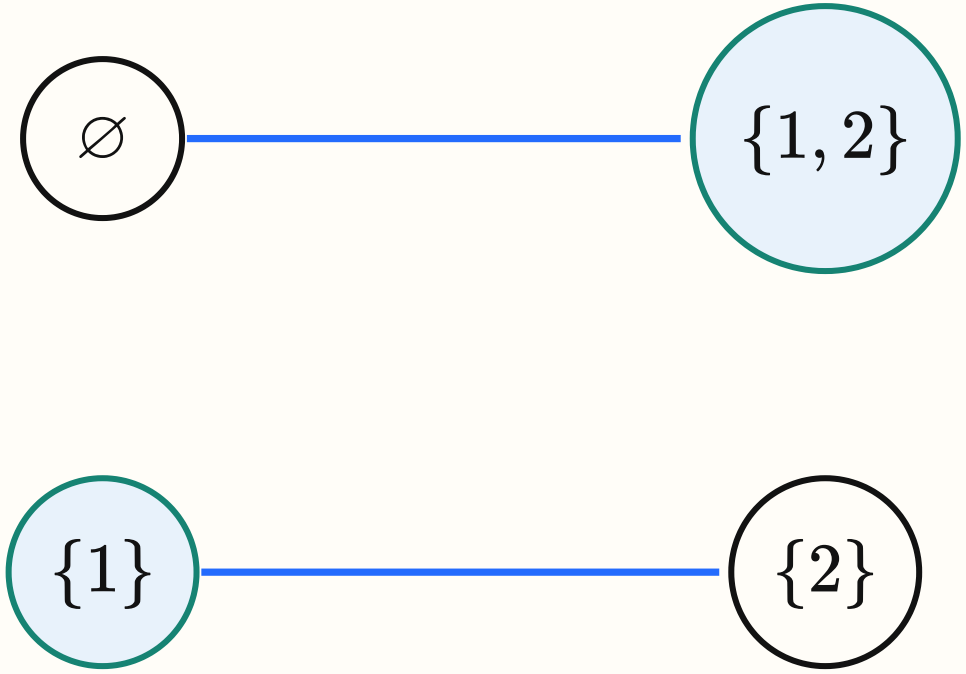
At most one per pair

$A \cap (U \setminus A) = \varnothing$, so $k \leq 2^{n-1}$.

Achieve equality

Select every subset containing 1; choose the other $n - 1$ elements freely.

Example: $U = \{1, 2\}$



Lines pair complements, not order comparisons. Highlighted: all subsets containing 1.

TAKEAWAY

Order can leave pairs incomparable

Incomparability is different from equality: 8 and 12 are distinct maximal elements under divisibility.

